

CULVERTCRAWL · DESIGN DECISION RECORD

CulvertCrawl recommendations accepted

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DOCUMENT CONTROL

Document ID	CVC-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/culvertcrawl		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 4 to 6 decided by Amish on 2026-10-02 (CVC-DEC-001)
0.2	2026-09-26	Amish Chadha	Budget set to \$910 to cover the priced BOM: decided by Amish, 2026-09-26; item 7 closed; R12 met
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 3; item 7 decided 2026-09-26; items 4 to 6 decided 2026-10-02)

■ Context

The TRL 3 review (docs/REVIEW.md , session 2026-09-25, TRL 3) left six items awaiting Amish. Three had a recommendation: R2 reach, R5 accuracy at 900 mm and R12 cost. Three did not: the first output format, the first partner user group and the removable laser boom. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos."

This record lists what that instruction decides, what changed in the repo, and what stays open. TRL 4 is on hold by Amish's instruction, so every decision is applied at TRL 3 (paper design, model, drawing and calculation) only.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 3 section, "Still awaiting Amish", items 4 to 6) and in CVC-CAL-001 v0.1. Where several options were offered, the recommended option is the decision.

■ Decision

Table 1. Items decided on 2026-09-25.

#	Item	Decision	What changed in the repo
1	R2 reach in the wet, uphill design case (was 39 m against 50 m)	Decided by Amish, 2026-09-25: go with recommendation. (a) Add about 0.40 kg of ballast, plus (b) entering flooded pipes from the upstream end as operating guidance	Ballast plate enlarged from 210 x 80 x 12 mm (1.67 kg) to 230 x 80 x 14 mm (2.13 kg), set 8 mm above the track contact line (was 10 mm): cad/src/model.py , STEP and STL, bom/bom.csv line 8 (\$15 to \$18). Crawler 5.31 to 5.77 kg; net submerged 2.66 to 3.05 kg. Wet uphill reach 39 to 52 m; dry uphill 52 to 56 m. Upstream entry added as operating guidance in CVC-PRC-001 v0.4 and CVC-REQ-001 v0.4. Measuring submerged track friction is TRL 4 work: decided, on hold
2	R5 accuracy at 900 mm (was 1.59 %)	Decided by Amish, 2026-09-25: go with recommendation. (a) Move the laser ring plane from 200 to 300 mm ahead of the camera	ring_d 200 to 300 mm in cad/src/model.py ; boom 100 mm longer; crawler length 510 to 610 mm; bom/bom.csv line 7 (\$45 to \$47). Vertical diameter error at 900 mm 1.59 to 0.97 %; at 600 mm 0.87 to 0.62 %; at 300 mm 0.35 to 0.24 %. Calibration in reference pipes is TRL 4 work: decided, on hold
3	R12 cost (\$901 against \$900)	Decided by Amish, 2026-09-25: go with recommendation. Accept the \$1 overrun as within pricing uncertainty and revisit when real quotes exist	budget_usd stays 900; R12 target text unchanged; R12 status records the accepted \$1. Getting real quotes is purchasing work at TRL 4: decided, on hold

Knock-on changes, all at TRL 3:

- CVC-CAL-001 v0.1 to v0.2: results rerun against the new model and BOM; the profiling Monte Carlo now uses 2,000 trials instead of 400 so that the 95th percentile at 900 mm is stable to about 0.03 percentage points.
- Drawing CVC-DWG-001 Rev P1 to Rev P2 (ring plane, ballast, length and mass notes).
- Concept media regenerated from the model (`media/`).
- CVC-PRB-001, CVC-PRC-001 and CVC-REQ-001 v0.3 to v0.4; CVC-DDR-001 v0.1 to v0.2; `README.md` .
- `project.yaml` : unchanged (`budget_usd`: 900 , `trl`: 3 , `trl_target`: 3 ; pitch and problem wording unchanged).

Items that remain open

These stayed open; Amish approved the recommendations written for them in CVC-DEC-001 on 2026-10-02 ("i approve your recommendations for all 555 open decisions."):

1. **Output format first** (PACP-style observation log, FHWA condition ratings or a deflection report). No recommendation was made. Decided 2026-10-02: the deflection report first, PACP vocabulary second, FHWA ratings later.
2. **First partner user group** for field trials. Not named; partners are picked per area later. Decided 2026-10-02: a county road department, with a university transportation program as backup (first candidate to approach: the Texas A&M Transportation Institute).
3. **Removable laser boom** for tight or bent pipes. No recommendation was made. The question matters more now that the boom overhangs the tracks by about 300 mm. Decided 2026-10-02: two bolts at the fin bracket for the prototype; a tool-free clamp after the first pipe trials.
4. **New: the \$5 added by items 1 and 2.** The priced BOM is now \$906, \$6 over the \$900 budget. Decision 3 accepted only the original \$1. Recommendation: accept the \$906 total on the same basis (indicative prices, revisit with real quotes). `budget_usd` is not changed. Decided by Amish, 2026-09-26: see Budget, 2026-09-26.

Budget, 2026-09-26

On 2026-09-26 Amish wrote: "i approve all the budget items." Budget set to \$910 to cover the priced BOM: decided by Amish, 2026-09-26. The priced BOM is \$906 (`bom/bom.csv` , 13 lines), so `budget_usd` in `project.yaml` moves from 900 to 910, item 7 is closed and R12 moves from not met to met (\$4 under). `docs/04-calcs/sizing.py` reads the budget from `project.yaml` and was rerun. CVC-CAL-001 v0.3, CVC-REQ-001 v0.5, CVC-PRC-001 v0.5, CVC-PRB-001 v0.5, `README.md` , `bom/bom-notes.md` and the concept blueprint key figure were updated, and `media/` was regenerated. Requirement status is now 0 not met, 2 at risk (R7, R8), 1 not verifiable at TRL 3 (R6) and 10 met on paper.

■ Consequences

- R2 and R5 move from not met to met on paper, both with thin margins; R12 remains not met by \$6, of which \$1 is accepted.
- Requirement status (CVC-CAL-001 v0.2): 1 not met (R12), 2 at risk (R7, R8), 1 not verifiable at TRL 3 (R6), 9 met on paper (R1, R2, R3, R4, R5, R9, R10, R11, R13).
- The crawler is 0.46 kg heavier and 100 mm longer; kit mass rises from 17.6 to 18.1 kg (R10 still met), sprocket torque at the traction limit from 0.67 to 0.73 N·m (within the 1 N·m motor rating) and the locked-track recovery pull from 50 to 53 N (factor 18.9 on the 1 kN tether).

- No cross-repo action arises from these decisions.
- No TRL 4 work is started. `trl` and `trl_target` stay at 3.